Utrecht University

TU/e EINDHOVEN UNIVERSITY OF TECHNOLOGY

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# *The Educational Proof Assistant Waterproof in an Introductory Proof Course*

## *Proof Construction and Learning Processes*

**Pim Otte, Rogier Bos, Johan Commelin, Jim Portegies**

ThEdu 2026

# Outline

Motivation

Context

Methods

Results

Conclusion

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  - Immediate feedback.
  - Transferable to paper.

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- Related: Mizar, Isar, Lurch, Naproche, Diproche, Verbose Lean, Yalep, Proofweb, ProofBuddy, Deaduction, Edukera, . . .

# Related educational work

- Learning about Proof with the Theorem Prover LEAN: the Abundant Numbers Task [Thoma and Iannone, 2022]
- Comparative analysis of student proof structure following the use of Waterproof [Wemmenhove et al., 2026].

# Demo

`https://impermeable.github.io/aquarium/`

# Research questions

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3. Surveys, observation, proof structure analysis, grades.

# Participants

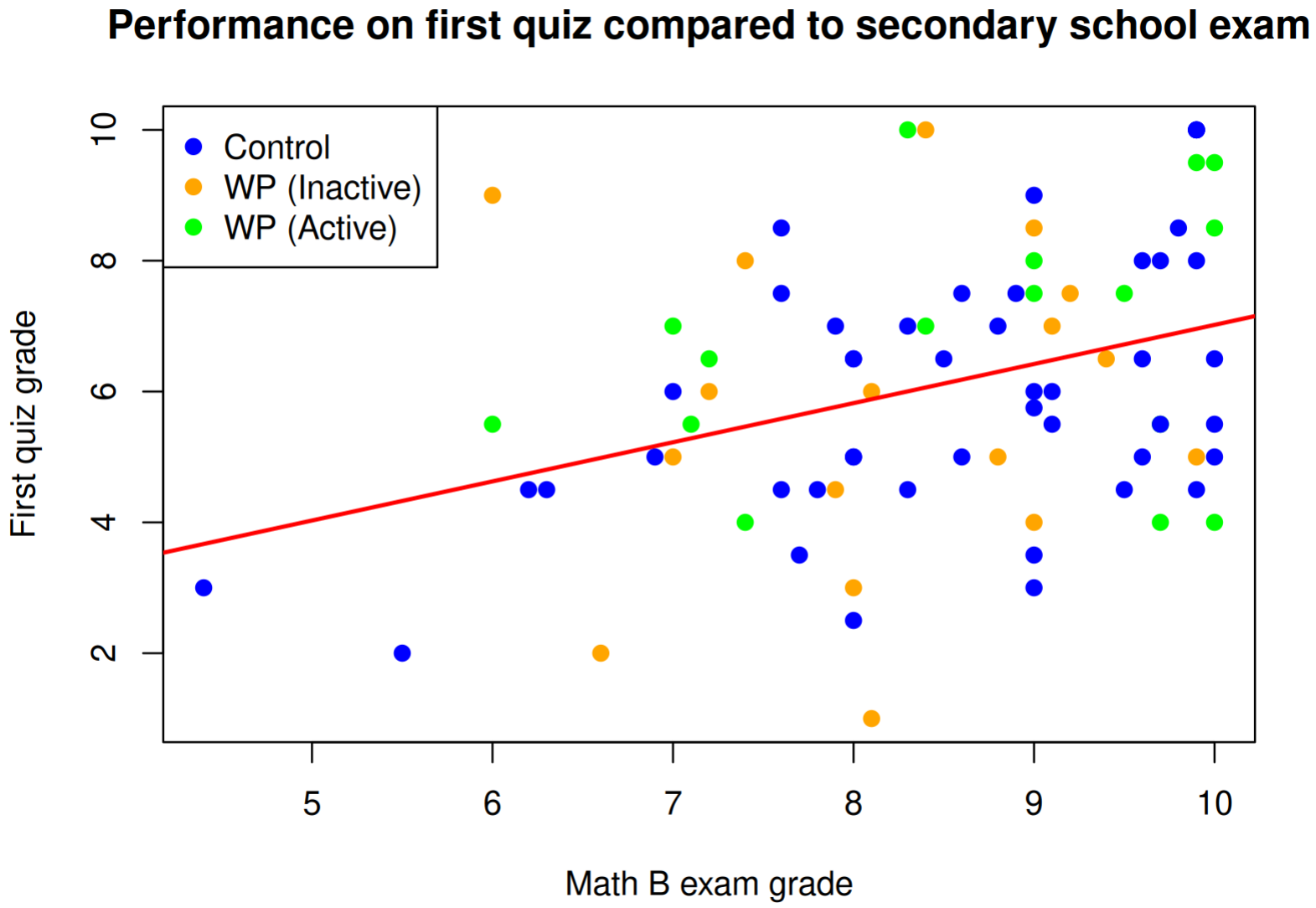
| Study Programme | Waterproof | Control |
|---|---|---|
| Mathematics (no double bachelor) | 7 | 14 |
| Mathematics and applications | 0 | 9 |
| Mathematics/Computer Science | 13 | 0 |
| Mathematics/Economics | 1 | 5 |
| Mathematics/Physics | 9 | 14 |
| Other: | 4 | 4 |

# Performance on first quiz

Table: Mean Quiz 1 results by study programme and condition

| Study Programme | WP (active) | WP (inactive) | Control |
|---|---|---|---|
| Mathematics (no double bachelor) | 6.12 | 5.75 | 5.96 |
| Mathematics and applications | | | 4.61 |
| Mathematics/Computer Science | 8.00 | 6.78 | |
| Mathematics/Economics | 5.50 | | 6.65 |
| Mathematics/Physics | 7.20 | 4.12 | 6.75 |
| Other: | 5.75 | 4.50 | 5.62 |

# Performance vs Math B grade



# Proof analysis

Table: Part of analyzed proof of participant 11 (no indentation)

| **Proof step** | **Code** |
|---|---|
| ... | ... |
| - Case q(14). | signpost-case |
| Assume that q(14). | continued |
| Choose y = 14. | ∃-intro |
| {Indeed, $14 \in \mathbb{R}$.} | domain-check |
| It holds that $\exists y \in \mathbb{R},\ q(y)$. | continued |
| We conclude that $(\neg p(x)) \vee (\exists y \in \mathbb{R},\ q(y))$. | signpost-concl |
| - Case $\neg p(x)$. | signpost-case |
| Assume that $\neg p(x)$. | continued |
| It holds that $\neg p(x)$. | continued |
| We conclude that $(\neg p(x)) \vee (\exists y \in \mathbb{R},\ q(y))$. | or-intro,signpost-concl |
| End of proof. | continued |

# Difference in proof steps

Table: Average occurrences of selected logical steps per student, by group

| Code | Control | WP (Inactive) | WP (Active) |
|---|---|---|---|
| or-elim | 0.70 | 0.65 | 1.44 |
| signpost-case | 2.78 | 3.12 | 4.31 |
| exists-intro | 1.46 | 1.41 | 2.31 |

# Transfer across language

| |
|---|
| Te bewijzen is dat $\forall y \in f(A), \exists x \in A, y = g(x)$. |
| Neem $y \in f(A)$. |
| $f(A) = \{f(x) \mid x \in A\}$, dus $\exists x' \in A, y = f(x)$. |
| Verkrijg deze $x' \in A$. |
| Er geldt dus $y \in f(x')$. |
| We moeten bewijzen dat $\exists x \in A, y = g(x)$. |
| Kies $x := x'$. |
| (Inderdaad, $x \in A$.) |
| We moeten bewijzen dat $y = g(x')$. |
| Per definitie van $g$ geldt dat $g(x') = f(x')$, |
| dus het is waar dat $y = g(x')$. |
| We concluderen dat $g$ surjectief is. |

Table: Dutch proof of surjectivity of a function given on the exam by an active Waterproof user, showing a Waterproof-inspired proof

# Selected survey responses

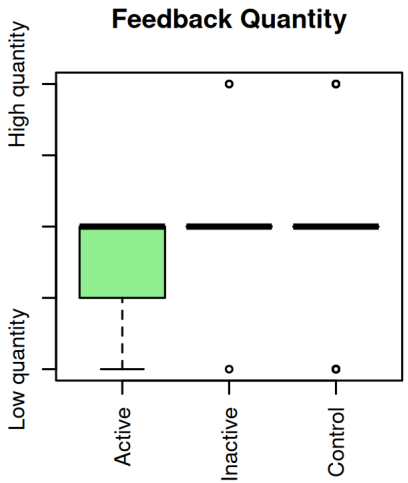
- “It helped me structure my answers, it taught me the basics on how to tackle problems, but it stopped me often when i tried to do something creative, because it was quite strict.”
- “The ‘by magic’ command is really fun”
- “While I do think Waterproof has definitely helped me with more quickly developing my proving skills, the actual program itself is still a bit buggy/slow”
- “I think it’s an excellent tool for learning proofs in mathematics, however I do think it needs to be accompanied by a textbook or lessons alongside it”

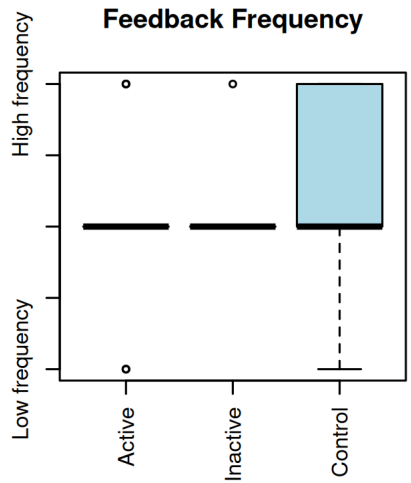
# Conclusion

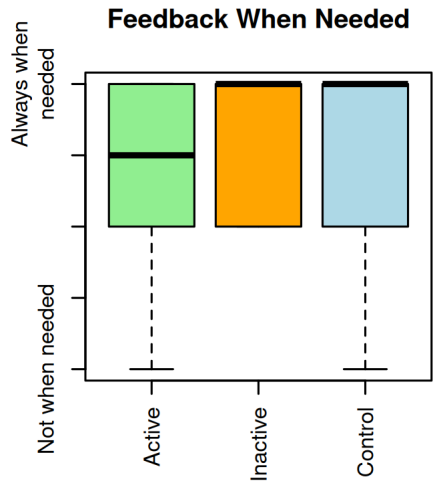
- Waterproof helps students to make their proofs more explicit.
- Waterproof might be extra helpful for Computer Science double majors.
- There is still room to improve Waterproof.

# Thanks for your attention!

# Perceptions about feedback

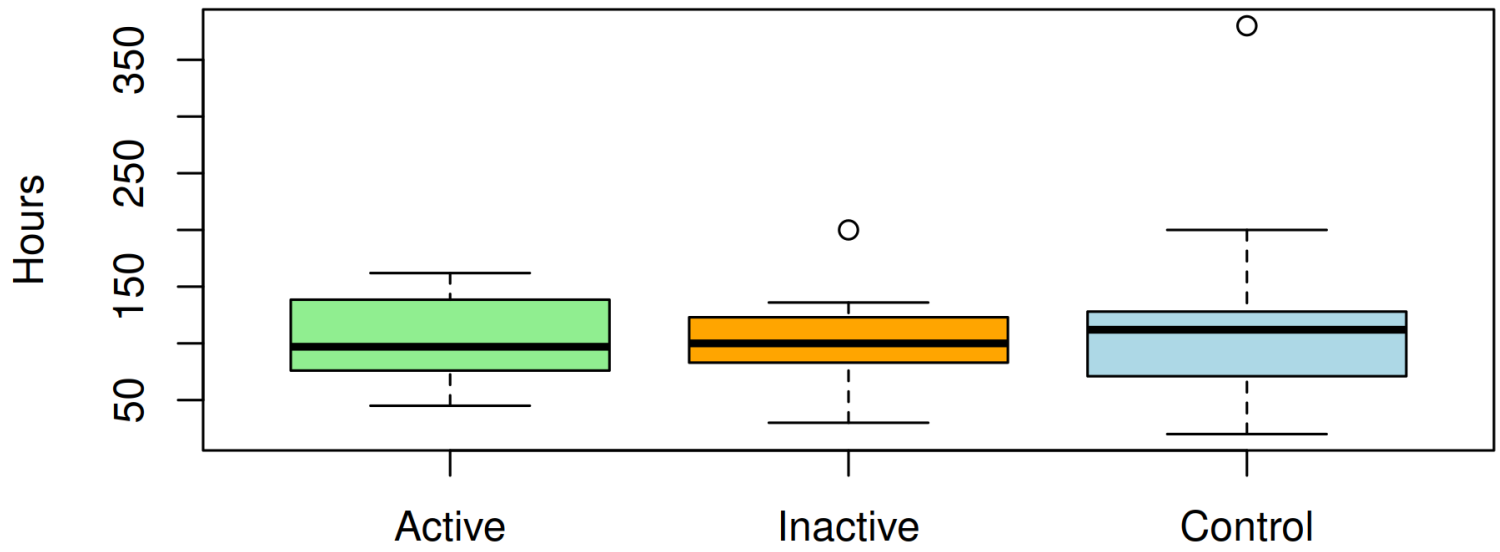


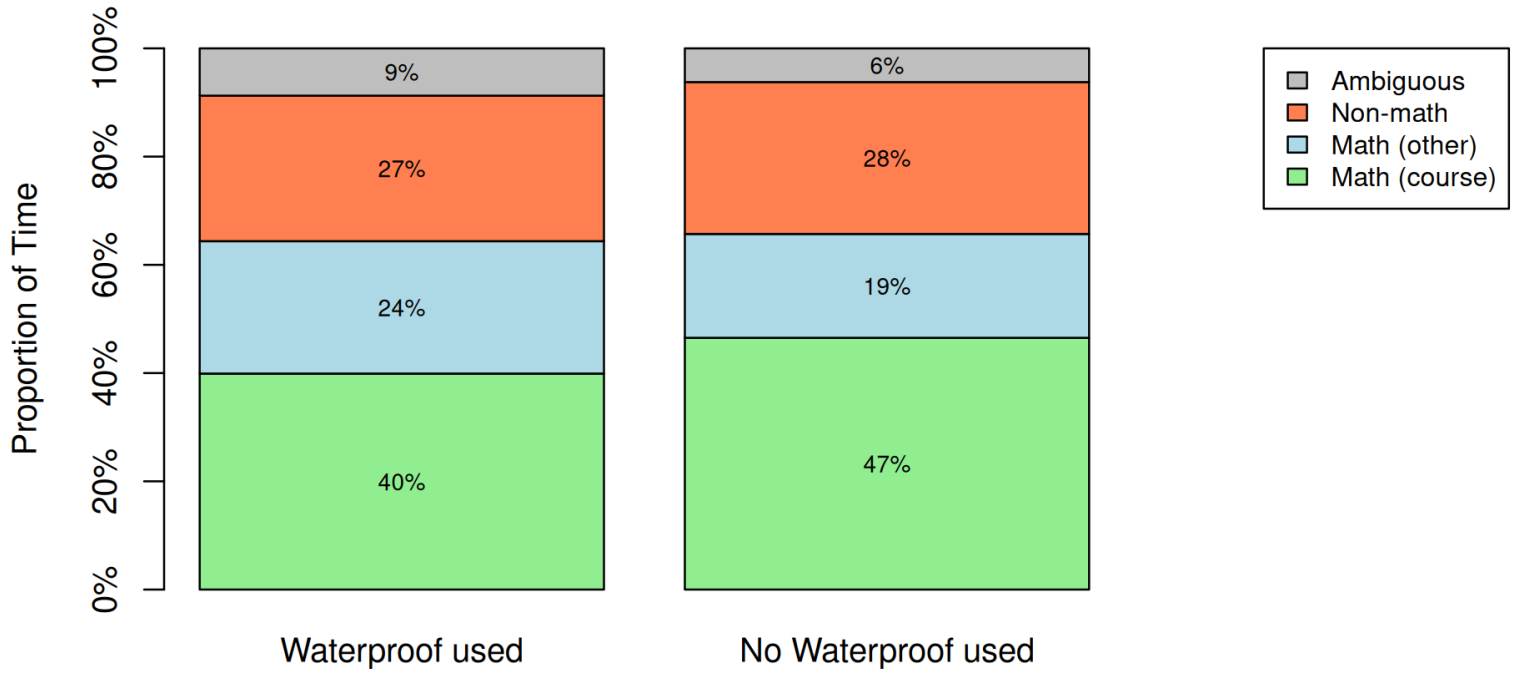

Survey results on perceptions about feedback

# Time spent on the course



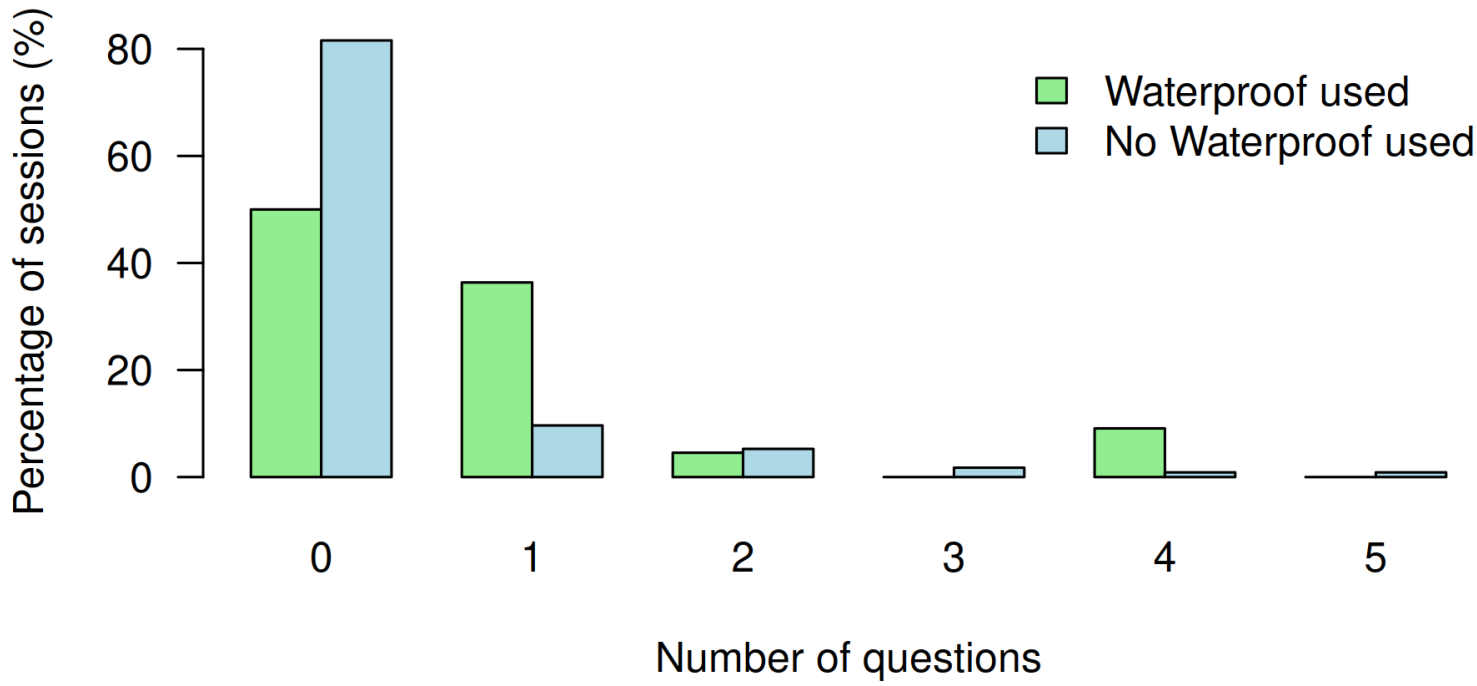

Survey results on time spent on the course

# Time on task (in class)



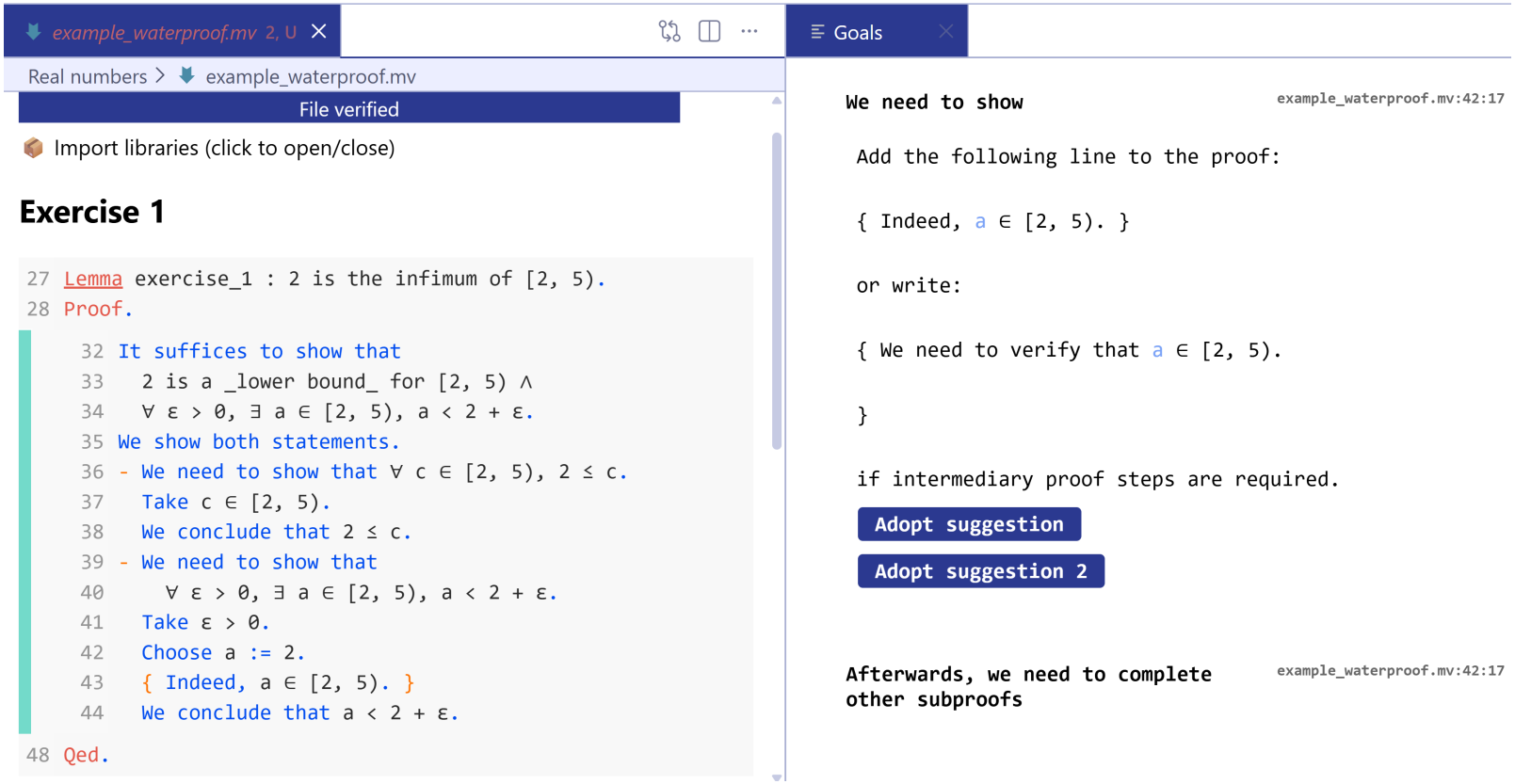

Proportion of observed in-class time spent in each activity category, averaged over sessions and grouped by whether Waterproof was used.

# Help-seeking behavior




Observations on number of questions asked by students

# Waterproof screenshot




A Waterproof proof and example of feedback.

# References I


Pim Otte, Rogier Bos, Johan Commelin, and Jim Portegies. The educational proof assistant waterproof in an introductory proof course: Proof construction and learning processes, 2026. URL https://arxiv.org/abs/2606.26809.

Athina Thoma and Paola Iannone. Learning about proof with the theorem prover lean: the abundant numbers task. *International Journal of Research in Undergraduate Mathematics Education*, 8(1):64–93, Apr 2022. ISSN 2198-9753. doi: 10.1007/s40753-021-00140-1. URL https://doi.org/10.1007/s40753-021-00140-1.

# References II


Jelle Wemmenhove, Dick Arends, Thijs Beurskens, Maitreyee Bhaid, Sean McCarren, Jan Moraal, Diego Rivera Garrido, David Tuin, Malcolm Vassallo, Pieter Wils, and Jim Portegies. Waterproof: Educational software for learning how to write mathematical proofs. In Julien Narboux, Walther Neuper, and Pedro Quaresma, editors, *Proceedings 12th International Workshop on Theorem proving components for Educational software. (ThEdu'23)*, 2024.

Jelle Wemmenhove, Nick Hoofd, Alexander Schüler-Meyer, and Jim Portegies. Comparative analysis of student proof structure following the use of waterproof. To appear in post-proceedings of ThEdu 25, 2026.



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pim.otte.dev